

An Internship Report
Of
AI-ML Virtual Internship

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: _____

Name of Student: ROHIT KUMAR

Batch: _____

CANDIDATE’S DECLARATION

I, ROHIT KUMAR, hereby declare that the internship report titled “AI-ML Virtual Internship” has been prepared by me in fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The internship was undertaken under the AICTE – EduSkills Virtual Internship Program from January 2026 to March 2026.

All sources of information used in this report have been acknowledged in the references. I affirm that this report is the result of my own work and has not been submitted to any other institute for any degree or diploma. I shall be solely responsible for any copyright violation in this regard.

Signature: _____

Student Name: ROHIT KUMAR

Roll No.: _____

ACKNOWLEDGEMENT

I take this opportunity to thank everyone who supported me during the internship. I am grateful for the guidance, constructive feedback and friendly advice that helped me complete this work, and for the honest views shared with me on many aspects of the subject.

I sincerely thank the AICTE – EduSkills Virtual Internship Program team, EduSkills, and Google for Developers for giving me the opportunity and a well-structured platform to gain hands-on exposure to Artificial Intelligence and Machine Learning. I am also thankful to my institute, IILM University, Greater Noida, and to the faculty of the School of Computer Science and Engineering for their continued support and encouragement.

Finally, I thank my family and friends for their constant encouragement throughout this journey.

Signature: _____

Student Name: ROHIT KUMAR

Roll No.: _____

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3. INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

This report summarises the 10-week Virtual Internship in Artificial Intelligence and Machine Learning (AI-ML) that I completed under the AICTE – EduSkills Virtual Internship Program, with support from Google for Developers through the India Edu Program. The internship ran from January 2026 to March 2026 and forms part of the requirements of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The purpose of the internship was to build a practical foundation in machine learning and deep learning with TensorFlow, and to apply it to computer-vision tasks such as image classification, on-device object detection and product image search for mobile applications. The programme followed a week-wise plan that combined concept learning with hands-on labs, quizzes and a final assessment.

After completing all requirements, I was awarded a Certificate of Virtual Internship with grade O (Outstanding), issued by EduSkills in association with the Ministry of Education and AICTE. The certificate is reproduced in Chapter 3 of this report.

4.2 Organization Profile

EduSkills works under the theme “Nation Building Through Skills”. It runs the AICTE – EduSkills Virtual Internship Program in association with the All India Council for Technical Education (AICTE) and the Ministry of Education, Government of India, and the programme is offered through the AICTE National Internship Portal.

AICTE is the national body responsible for the planned and coordinated development of technical education in India. For the AI-ML track, Google for Developers supports the programme through its India Edu Program, as stated on the certificate. Together, these organisations aim to give engineering students structured, industry-relevant exposure to machine learning.

4.3 Problem Statement

Most engineering students meet machine learning and neural networks only in theory during their coursework. Few get structured practice in building, training and deploying computer-vision models with an industry-standard framework such as TensorFlow. Ideas such as convolutional neural networks, object detection and image classification often stay abstract because students rarely train a model, convert it for a phone, and wire it into a real application.

This internship addressed that gap. Its aim was to learn the complete TensorFlow workflow in practice: building and training a neural network, detecting and classifying objects on a device, and creating a product image search capability for a mobile app.

4.4 Project Objectives

- To understand the fundamentals of machine learning and build convolutional neural networks (CNNs) with TensorFlow for image recognition and classification.
- To learn the basics of object detection and integrate pretrained detectors into mobile applications.
- To train custom object-detection models using TensorFlow Lite and Model Maker.
- To build a product image search feature based on on-device object detection.
- To detect objects in static images and live camera feeds for visual product search.
- To integrate a backend service with a product image search mobile application.
- To build custom image-classification models and improve their accuracy.
- To create a custom image classifier and integrate it into an application.
- To clear the final credential validation and earn the internship grade.

4.5 Scope of the Project

The internship covered the TensorFlow-based computer-vision workflow from end to end. It began with neural-network fundamentals and CNNs, moved to on-device object detection with pretrained and custom-trained models, and then to a product image search feature that combines detection with visual search across still images and live camera feeds. It also covered building and improving custom image classifiers and integrating them into an application.

The work was carried out in a virtual, lab-based learning environment. It did not involve deployment on any live production system of an organisation.

4.6 Technologies and Tools Used

- **Programming and ML framework:** Python, TensorFlow, Keras
- **On-device machine learning:** TensorFlow Lite and Model Maker
- **Computer vision:** Convolutional Neural Networks (CNNs), pretrained and custom object detectors, custom image classifiers
- **Application layer:** Mobile application integration and camera-feed based live object detection
- **Learning platform:** Online learning pathways, quizzes and skill badges provided through the programme

4.7 System Architecture

The learning track can be understood as a set of connected layers, each building on the one before it:

- **Model layer:** A CNN built and trained in TensorFlow provides the core image-recognition capability.
- **Conversion layer:** Trained models are converted into a lightweight format for phones using TensorFlow Lite and Model Maker.
- **Detection layer:** Pretrained and custom detectors find and locate objects in still images and live camera feeds.
- **Search layer:** Product image search combines on-device detection with a backend service to enable visual search.
- **Classification layer:** A custom image classifier categorises detected objects and images more accurately.
- **Application layer:** The converted models are integrated into a mobile application that uses detection, search and classification.
- **Assessment layer:** Quizzes, skill badges and the final credential validation track progress and decide the grade.

4.8 Methodology

The internship followed a week-wise plan that combined conceptual learning, hands-on labs and quiz-based assessment, and ended with a final credential validation. The modules are summarised in the table below.

Week	Module	Description
1	Program Neural Networks with TensorFlow	Machine-learning fundamentals from first principles, leading to convolutional neural networks (CNNs) for image recognition and classification.
2	Get Started with Object Detection	Core ideas of object detection and how to add a pretrained detector to a mobile application.
3	Go Further with Object Detection	Training a custom object-detection model with TensorFlow Lite and Model Maker.
4	Get Started with Product Image Search	Building a product image search feature that relies on on-device object detection.
5	Get Started with Product Image Search	Detecting objects in still images and in live camera feeds to support visual search.
6	Go Further with Product Image Search	Connecting the product image search app to a backend service.
7	Go Further with Image Classification	Building custom image-classification models and improving their accuracy.
8	Go Further with Image Classification	Creating a custom image classifier and integrating it into an application.
9	Skill Badges and Revision	Completing remaining labs and quizzes, earning skill badges, and revising the key concepts.
10	Final Credential Validation	Final assessment used to determine the internship grade.

Each week was followed by quiz-based assessment to check conceptual understanding. In the final phase, a final credential validation was conducted, and the overall grade reflected performance across the weekly modules and this final assessment.

4.9 Expected Outcomes

- Ability to explain and apply the fundamentals of machine learning to build CNNs with TensorFlow.
- Practical understanding of object detection, including pretrained detectors and custom TensorFlow Lite models.
- Hands-on exposure to building a product image search feature that combines on-device detection with backend integration.
- Ability to build, evaluate and improve custom image-classification models.
- Skill in integrating converted TensorFlow Lite models into a mobile application.

- Completion of the weekly modules and the final credential validation, resulting in grade O (Outstanding).

4.10 Certificates of Completion

The Certificate of Virtual Internship issued by EduSkills, confirming successful completion of the 10-week AI-ML Virtual Internship during January – March 2026, is attached in Chapter 3 of this report. The certificate carries Certificate ID 408fba6b9b6f8410ffbb6, and the grade shown on its seal is O (Outstanding). It also carries a QR code that can be scanned to verify it.

5. BIBLIOGRAPHY/REFERENCES

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